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# Pneumonia associated with *Mycoplasma* spp in three cats

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*Mycoplasma* spp were isolated in pure culture from bronchoalveolar lavage specimens from three cats with clinical, cytological and radiographic signs of bronchopneumonia or suppurative bronchitis. Predisposing factors were not identified in the first case, the second cat had oesophageal hypomotility, while the third cat had been exposed to cigarette smoke and had advanced periodontal disease. Respiratory signs resolved promptly and completely in all cases following antimicrobial therapy directed against mycoplasmas. *Mycoplasma* spp are possible causes of lower respiratory tract disease in cats and this should be considered when selecting empirical therapy for feline airway disease and pneumonia. In some situations mycoplasmas may behave as primary lower respiratory tract pathogens in cats.

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Key words: *Mycoplasma* spp, pneumonia, cats.

BAL      Bronchoalveolar lavage  
PO       Orally

The role of mycoplasmas in feline pulmonary disease is poorly defined. Early studies reported conflicting data on recovery of mycoplasmas from the lower respiratory tract of healthy and diseased cats.<sup>1-4</sup> More recent work has indicated that recovery of mycoplasmas from tracheo-bronchial lavage specimens is an abnormal finding<sup>5,6</sup> but uncertainty exists as to whether mycoplasmas isolated from cats with lower respiratory tract disease are primary pathogens, opportunistic invaders or merely colonisers of abnormal sites. Certainly there are very few reports documenting diagnosis and treatment of mycoplasmal pneumonia in cats.

This report describes three cats in which chronic mycoplasmal bronchitis or bronchopneumonia was diagnosed on the basis of clinical signs, radiographic findings, pure culture of *Mycoplasma* spp from BAL specimens and prompt response to treatment with antimicrobials effective against mycoplasmas.

## Case 1

A 7-month-old spayed female Spotted Mist cat (4.2 kg) was referred with a 3 month history of paroxysmal coughing. During these episodes, the cat was seen to crouch, extend her neck, cough repeatedly, then breathe rapidly and shallowly. She had been vaccinated against feline herpes-, calici-, parvo- and leukaemia viruses and had no other history of ill health. Abnormal physical findings included pyrexia (39.3°C), mild mandibular lymphadenomegaly and wheezes on thoracic auscultation. Gentle tracheal palpation induced severe coughing accompanied by respiratory distress.

Haematological and serum biochemical findings were unremarkable. Thoracic radiography demonstrated a diffuse bronchoalveolar pattern and consolidation of the left cranial lung lobe (Figures 1 and 2). Unguided BAL was performed under light general anaesthesia using a sterile urinary catheter (10 French) passed through a sterile cuffed endotracheal tube and gently wedged in a bronchus of a caudal lung lobe. Diff Quik stained smears of the lavage specimen showed large numbers of neutrophils and smaller numbers of eosinophils, alveolar macrophages and lymphocytes. No

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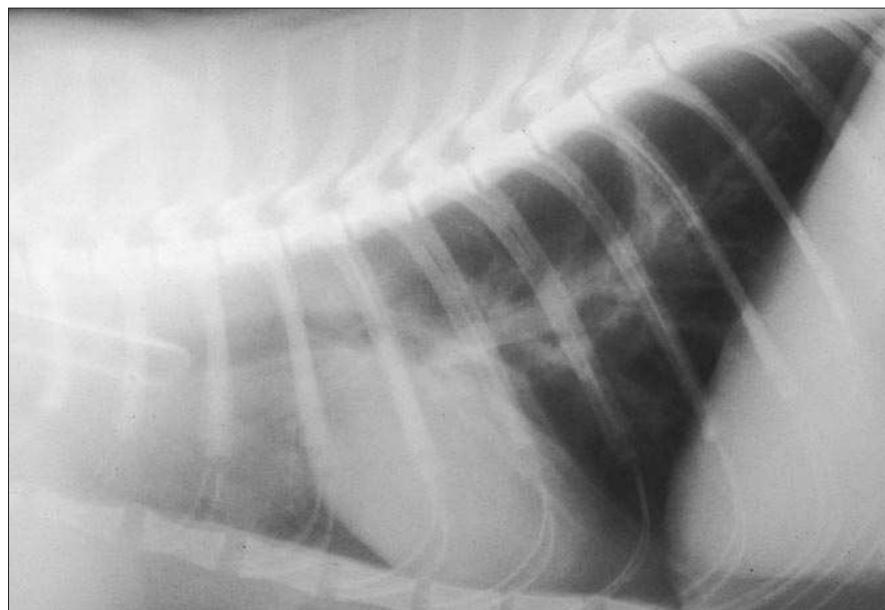
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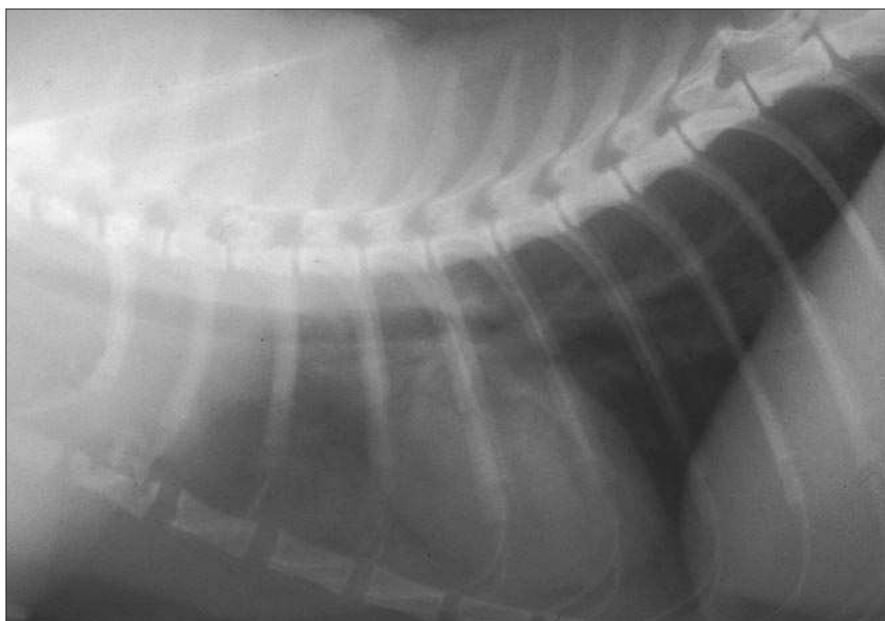
**Figure 1.** Ventrodorsal thoracic radiograph of a cat (case 1) before treatment. Note the consolidation of the left cranial lung lobe.

microorganisms were seen in smears stained using Burke's modification of the Gram-stain, viewed under light microscopy. A portion of the BAL specimen was inoculated onto 5% v/v sheep blood agar plates (Oxoid Blood Agar Base number 2) and incubated aerobically and anaerobically at 37°C. After 96 h, a heavy growth of a single colony type was observed. Colonies were approximately 0.5 mm diameter, circular, translucent grey with an entire edge and surrounded by a zone of complete haemolysis. Gram-staining of colonial material was difficult as the colonies were embedded in the agar, but no obvious bacterial structures were visible by light microscopy with oil immersion (x 1000). The morphological features of the colony and lack of Gram-staining were considered typical of a *Mycoplasma* spp and no further identification procedures were undertaken. Mycoplasmal bronchopneumonia was diagnosed on the basis of these findings.

When discharged, the cat was prescribed doxycycline (25 mg PO every 12 h for 6 weeks). The cough



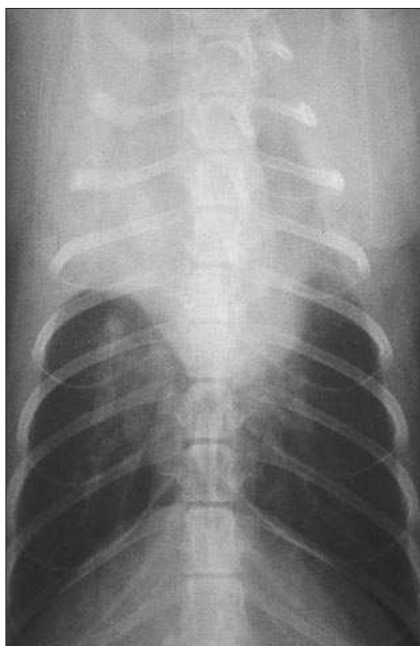
**Figure 2.** Right lateral thoracic radiograph of a cat (case 1) before treatment. Note the consolidation in the cranial lung lobe area. An endotracheal tube is also evident.



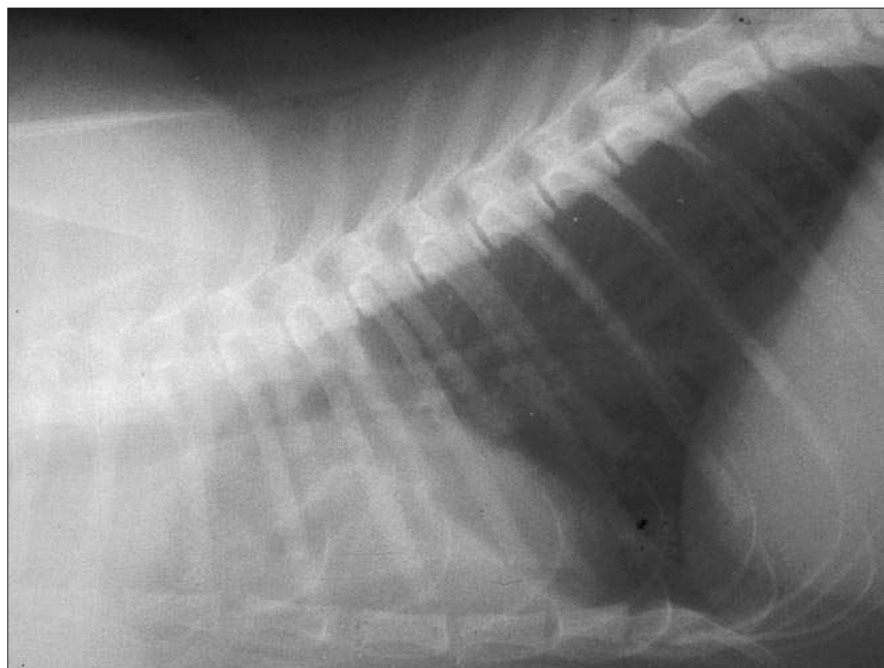
**Figure 3.** Right lateral thoracic radiograph of the cat in case 1 after 6 weeks therapy. Note improved aeration of the cranial lung lobe.

resolved within 1 week, the cat's general condition and demeanour also improved and she gained 0.6 kg in body weight. Radiographs taken after 6 weeks of treatment showed substantial improvement, although a bronchial pattern was still evident and there were some remaining areas of consolidation in the caudal part of the left cranial lung lobe, the caudal

portion of the right cranial lung lobe and the right middle lung lobe (Figure 3). A further 2 weeks of doxycycline was therefore prescribed. The cat was clinically normal for 18 months, when an intermittent cough was noted by its owners. Thoracic radiographs and BAL cytological results were suggestive of mild allergic bronchial disease. Conventional bacteria and myco-



**Figure 4.** Ventrodorsal radiograph of the cat in case 2 before treatment. Note the consolidation of the right cranial and middle lung lobes, and the left cranial lung lobe.



**Figure 5.** Right lateral radiograph of the cat in case 2 before treatment, showing consolidation of the cranial and middle lung lobe areas.

plasmas were not isolated from the lavage fluid. The cough resolved spontaneously.

### Case 2

A 6-month-old spayed female Burmese cat (2.6 kg) was referred with a history of chronic coughing, sneezing and bilateral ocular discharge. These signs became evident before weaning and worsened after ovariohysterectomy. The cough resolved whenever a course of amoxicillin-clavulanate was administered but recurred subsequently, while sneezing and ocular discharge persisted throughout. The cat also 'brought up' food two to three times weekly and the owner's description of these events suggested regurgitation. The cat had been vaccinated against feline calici-, herpes- and parvo-viruses.

Physical findings included pyrexia (39.4°C), an enlarged right popliteal lymph node and a moist cough elicited by gentle tracheal palpation. Results from thoracic auscultation were unremarkable. Thoracic radiographs demonstrated consolidation of the right cranial and middle lung lobes and the left cranial lung lobe (Figures 4 and 5). Unguided BAL was performed as for case 1. Cytological analysis of the lavage

fluid showed numerous neutrophils, occasional macrophages and respiratory epithelial cells. No microorganisms were seen in Gram-stained smears. The fluid was cultured aerobically and anaerobically on 5% v/v sheep blood agar and under increased atmospheric concentrations of carbon dioxide (10%) on chocolate agar. After 96 h incubation, a heavy pure growth of organisms with colony morphological characteristics of mycoplasmas was evident.

Pending culture results, empirical broad-spectrum antimicrobial therapy was commenced using ciprofloxacin (62.5 mg PO every 12 h) and amoxicillin (50 mg PO every 12 h). In addition, the kitten's airways were nebulised with saline for 5 to 10 min twice daily for 3 weeks. The owners declined investigation of the regurgitation. Antimicrobial therapy was continued with ciprofloxacin alone once culture results were known. At recheck 3 weeks later there was much improvement in respiratory signs but regurgitation persisted. Examination with barium fluoroscopy demonstrated oesophageal hypomotility and mild megaesophagus. The prokinetic drug cisapride (5 mg PO every 12 h) was prescribed in an attempt to improve oesophageal motility and

gastric emptying. The owners were advised to feed the cat with its upper body elevated. Six weeks after initial presentation, the kitten no longer coughed and repeat radiographs showed improved aeration of the left cranial lobe and only localised alveolar densities in the right cranial and right middle lung lobes.

Treatment with ciprofloxacin and cisapride was continued for a further 3 weeks, then the owners discontinued treatment against advice. Respiratory signs were no longer obvious and the frequency of regurgitation had decreased substantially. At the time of writing (2 years later), the owners report the cat coughs very rarely and regurgitates occasionally (once every 1 to 2 weeks). The owners declined further examination and radiography.

### Case 3

A 10-year-old castrated male domestic short-haired cat was presented with a 3 to 4 month history of coughing, weight loss and general unthriftiness. The cough had become progressively worse. For 2 weeks prior to admission the cat had also begun to dribble saliva and protrude its tongue. Its owner was a heavy smoker. Physical findings



included sialosis, severe periodontal disease with many loose and painful teeth, wastage of hindlimb musculature and open-mouth breathing during handling or exertion. Increased breath sounds on inspiration were noted during thoracic auscultation and the cat sneezed occasionally.

Thoracic radiographs taken under sedation demonstrated a severe diffuse bronchial pattern. The cat was then anaesthetised and an unguided BAL performed. Even before instillation of saline, the cat coughed copious quantities of yellow purulent fluid up through the endotracheal tube. Specimens of this fluid and BAL samples collected subsequently were submitted to the laboratory. Numerous loose teeth were extracted, and remaining teeth scaled and polished. Amoxycillin-clavulanate was administered at the completion of these procedures, and the cat discharged from hospital the next day and prescribed doxycycline (25 mg PO every 12 h).

Diff Quik stained smears of BAL specimens demonstrated numerous degenerate and intact inflammatory cells, consisting of neutrophils (more than 95% of the cells), macrophages and occasional lymphocytes. No microorganisms were seen in Gram-stained smears. A portion of BAL fluid was inoculated onto sheep blood agar and incubated aerobically and anaerobically. After 3 to 5 days, a moderate, pure growth of a *Mycoplasma* sp was observed.

The cat stopped coughing 48 h after discharge. Its appetite improved markedly and it gained 0.5 kg in 4 weeks. Its general condition also improved, the coat became lustrous and signs referable to periodontal disease, such as drooling and tongue protrusion, were no longer evident.

## Discussion

The pathogenic potential of mycoplasmas in cats has long been recognised but they are considered mainly in cases with upper respiratory and/or ocular disease.<sup>7</sup> Reports clearly documenting lower respiratory tract involvement in cats are few. There are anecdotal reports of mycoplasmal pneumonia in kittens, both naturally occurring<sup>8,9</sup> and experimental.<sup>10</sup> A radiograph of a cat with mycoplasmal pneumonia is

presented in a textbook.<sup>11</sup> Other cases have been diagnosed at the University of Bristol (P Galloway personal communication) and in cats immunosuppressed by glucocorticoids.<sup>9</sup> There are also sporadic reports of pyothorax and pulmonary abscessation involving mycoplasmas.<sup>12-14</sup>

The three cases presented here emphasise the potential for mycoplasmas to be associated with lower respiratory disease in cats. However, it remains unclear whether mycoplasmas behave as primary respiratory pathogens in cats, as they do in humans,<sup>15</sup> goats, pigs and cattle,<sup>16</sup> or as opportunistic invaders.<sup>17</sup> Mycoplasmas comprise part of the normal resident flora of the pharynx in many cats<sup>4,6,18,19</sup> and it is possible that tracheobronchial disease of another aetiology permits them to colonise the lower respiratory tract. Several experimental observations support the notion that airway colonisation may occur when there is concurrent disease of the airways or lung, typically of viral or allergic aetiology. No mycoplasmas were cultured from bronchoalveolar/tracheobronchial lavage specimens from 24 healthy cats<sup>5</sup> and, in another study of cats with and without pulmonary disease, mycoplasmas were only isolated from cats with lung disease.<sup>6</sup> Isolation of mycoplasmas from BAL fluid could reflect contamination of the sample during collection, but a heavy pure growth would be unlikely from such a sample and cytological evidence of pharyngeal contamination would also be likely.

Mycoplasmas clearly behaved as pathogens in the three cats reported here, as they were isolated in large numbers in pure growth from appropriate samples in which there was cytological evidence of inflammation, but no evidence of involvement with other microbes. Furthermore, antimicrobial therapy directed against them resulted in prompt resolution of the chronic respiratory signs. There were no known events predisposing the first cat to mycoplasmal pneumonia and it seems probable that in this animal mycoplasmas acted as a primary pathogen. Mycoplasmas may have also behaved as a primary pathogen in the second cat, although the findings could also be attributed to aspiration of ingesta and oropharyngeal contents secondary

to oesophageal dysfunction. Administration of amoxycillin-clavulanate may have then eliminated conventional bacteria, but not mycoplasmas, from the lower respiratory tract. In the third case, smoke inhalation and severe periodontal disease may have predisposed the cat to bronchitis, but the microbiological results and unequivocal response to treatment strongly argue that mycoplasmas were responsible for the cat's suppurative airway disease.

Although the organisms isolated in each case were not identified to the species level by biochemical reactions and testing with specific antisera at a reference laboratory, the lack of Gram-staining and the characteristic appearance of the colonies on blood agar were sufficient for their identification as *Mycoplasma* spp. It is generally considered that isolation of mycoplasmas is difficult and requires specialised media.<sup>9,20,21</sup> However, the present cases emphasise the ability to culture some species on nonspecific media containing peptone and yeast extract, such as the blood agar base used in our laboratory. Likewise, Gruffydd-Jones reported that mycoplasmas will grow on blood agar if the infection is heavy<sup>22</sup> and peptone-containing blood agar proved satisfactory for culturing a *Mycoplasma* sp from a kitten with pyothorax.<sup>14</sup> The present findings emphasise the importance of routinely culturing BAL specimens with inflammatory cytological features irrespective of whether bacteria are seen in Gram-stained smears. It would be prudent in such cases to use techniques favouring the recovery of mycoplasmas.

Antimicrobial susceptibility tests for mycoplasmas have not been standardised and are not widely available.<sup>21</sup> Mycoplasmas are reported to be sensitive to macrolides (erythromycin and tylosin), azalides (or 'new macrolides' – azithromycin and clarithromycin), lincosamides (lincomycin and clindamycin), tetracyclines (including doxycycline), chloramphenicol and fluoroquinolones (including enrofloxacin and ciprofloxacin).<sup>7,21</sup> When using empirical antimicrobial therapy in cats with lower respiratory tract signs, it may be prudent to consider an agent with activity against mycoplasmas as primary therapy or if therapy with a beta-lactam antibiotic has failed. If mycoplasmal pneumonia is documented by culture, treatment with

azithromycin should be considered in certain situations. This agent is very effective against mycoplasmas, accumulates inside phagocytes and has a dose rate (5 mg/kg PO once or twice a week) particularly suited to long-term dosing of cats.<sup>23,24</sup>

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## Addendum

The authors have recently seen a further case of feline mycoplasmal bronchopneumonia. The cat was a mature tom presented with a 2 to 3 month history of productive coughing. Chest radiographs demonstrated a severe diffuse bronchial pattern, with alveolar disease evident also in several lung lobes. BAL demonstrated much mucin, numerous non-degenerate neutrophils, and occasional macrophages and eosinophils. A moderately heavy pure growth of a *Mycoplasma* sp was observed after 2 to 3 days incubation on blood agar cultured aerobically. The cat's cough resolved and its weight increased from 4.0 to 4.5 kg during 3 weeks of doxycycline treatment (25 mg every 12 h), which was continued for an additional 4 weeks.